

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I **B SANTHOSH KUMAR** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B SANTHOSH KUMAR

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- li. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- lii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

Aim

To represent the given medical knowledge using propositional logic and apply Forward Chaining and Backward Chaining to obtain the possible diagnosis for Patient A.

(a) Representation using Propositional Logic

Step 1: Define the symbols

Symbol	Meaning
F	Patient A has Fever
C	Patient A has Cough
R	Patient A has Rash
B	Patient A has Breathlessness
Fl	Patient A has Possible Flu
M	Patient A has Possible Measles
P	Patient A has Possible Pneumonia

Step 2: Represent the Rules

Rule	Description	Propositional Form
Rule 1	If the patient has Fever and Cough, Possible Flu	$F \wedge C \rightarrow Fl$
Rule 2	If the patient has Fever and Rash, Possible Measles	$F \wedge R \rightarrow M$
Rule 3	If the patient has Cough and Breathlessness, Possible Pneumonia	$C \wedge B \rightarrow P$

Step 3: Represent the Given Facts

For Patient A, the given facts are:

Symbol	Fact	Value
F	Fever	True
C	Cough	True
B	Breathlessness	True
R	Rash	Not given

Knowledge Base

The knowledge base consists of the three rules and the given facts:

KB = {F, C, B, $F \wedge C \rightarrow FI$, $F \wedge R \rightarrow M$, $C \wedge B \rightarrow P$ }

Note: Rash is marked as “**Not given**” because the case

(b) Forward Chaining

Forward Chaining starts with the available facts and checks the rules whose conditions are satisfied.

Initial Facts

Symbol	Given Fact	Value
F	Fever	True
C	Cough	True
B	Breathlessness	True

Inference Table

Step	Conditions checked	Rule	Result
0	F, C, B are given	Initial facts	F, C, B
1	$F = \text{True}$ and $C = \text{True}$	R1: $F \wedge C \rightarrow FI$	$FI = \text{True}$
2	$C = \text{True}$ and $B = \text{True}$	R3: $C \wedge B \rightarrow P$	$P = \text{True}$
3	$F = \text{True}$ but R is not given	R2: $F \wedge R \rightarrow M$	M not derived

Working

Step	Rule Applied	Given Conditions	Derived Fact
1	Rule 1: $F \wedge C \rightarrow FI$	Fever (F) = True, Cough (C) = True	Possible Flu (FI) = True
2	Rule 3: $C \wedge B \rightarrow P$	Cough (C) = True, Breathlessness (B) = True	Possible Pneumonia (P) = True
3	Rule 2: $F \wedge R \rightarrow M$	Fever (F) = True, Rash (R) is not given	Possible Measles (M) is not derived

Result

Diagnosis	Result
Possible Flu	Derived
Possible Pneumonia	Derived
Possible Measles	Not derived

Reason: Rule 2 requires both Fever and Rash. Since the case gives no fact for Rash, the rule cannot be fired.

Forward Chaining Result

Diagnosis	Result
Possible Flu	True
Possible Measles	Not derived
Possible Pneumonia	True

(c) Backward Chaining for Pneumonia

Goal: Possible Pneumonia (P)

Rule 3 is: **Cough $\wedge$ Breathlessness $\rightarrow$ Possible Pneumonia**

To prove the goal **P**, the rule requires the following two conditions:

Sub-goal	Given fact	Status
Cough (C)	C = True	Satisfied
Breathlessness (B)	B = True	Satisfied

Since both sub-goals are satisfied, Rule 3 can be applied.

Rule	Conditions	Conclusion
Rule 3	Cough $\wedge$ Breathlessness	Possible Pneumonia

Result: Possible Pneumonia (P) is derived for Patient A

Goal Reduction

flowchart TD

A["Goal: P - Possible Pneumonia"] --> B["Rule 3: C $\wedge$ B $\rightarrow$ P"]

B --> C["Sub-goal: Cough (C)"]

B --> D["Sub-goal: Breathlessness (B)"]

C --> E["Fact: C = True"]

D --> F["Fact: B = True"]

E --> G["Both sub-goals satisfied"]

F --> G

G --> H["P = True"]

Verification

Step	Condition	Status
1	Cough (C)	Given fact – True
2	Breathlessness (B)	Given fact – True
3	$C \wedge B$	Both conditions satisfied
4	Rule 3: $C \wedge B \rightarrow P$	Rule can be fired
5	Possible Pneumonia (P)	Derived – True

Result

Patient A is identified as having **Possible Pneumonia** from the given knowledge base. Possible Flu is also derived, while Possible Measles is not derived.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

Aim

To apply First Order Logic, Unification and Resolution to determine whether CarA can proceed behind an emergency vehicle.

(a) Unification of Rule 1

Given Rule 1

$\forall x : \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Given Facts

Fact	Value
Vehicle(Ambulance)	True
EmergencyType(Ambulance)	True

Unification Steps

Step	Rule Expression	Given Fact	Substitution
1	Vehicle(x)	Vehicle(Ambulance)	$x = \text{Ambulance}$
2	EmergencyType(x)	EmergencyType(Ambulance)	$x = \text{Ambulance}$

Most General Unifier

$\theta = \{x/\text{Ambulance}\}$

After substitution:

Vehicle(Ambulance) $\wedge$ EmergencyType(Ambulance) $\rightarrow$ ClearPath(Ambulance)

Condition	Status
Vehicle(Ambulance)	True
EmergencyType(Ambulance)	True
ClearPath(Ambulance)	Derived

Result: ClearPath(Ambulance) is derived using Rule 1.

(b) Resolution Proof of Proceed(CarA)

Step 1 – Conversion of Rules into CNF

Rule	CNF Form
Rule 1	$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
Rule 2(a)	$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
Rule 2(b)	$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$
Rule 3	$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

For proving **Proceed(CarA)**, Rule 2(b) is not required.

Step 2 – Relevant Clauses

Clause	r
C1	$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
C2	$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
C3	$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$
C4	$\text{Vehicle}(\text{Ambulance})$
C5	$\text{EmergencyType}(\text{Ambulance})$
C6	$\text{Vehicle}(\text{CarA})$
C7	$\text{Behind}(\text{CarA}, \text{Ambulance})$
C8	$\neg \text{Proceed}(\text{CarA})$

C8 is the negation of the goal and is added for resolution refutation.

Step 3 – Derive ClearPath(Ambulance)

Step	Clauses Used	Resolvent
1	C1 + C4, $x = \text{Ambulance}$	$\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
2	Step 1 + C5	$\text{ClearPath}(\text{Ambulance})$

Step 4 – Derive GreenSignal(Ambulance)

Step	Clauses Used	Resolvent
3	C2, $x = \text{Ambulance}$ + $\text{ClearPath}(\text{Ambulance})$	$\text{GreenSignal}(\text{Ambulance})$

Step 5 – Derive Proceed(CarA)

Substitute:

x = CarA

y = Ambulance

Rule 3 becomes:

$\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

Given:

Condition	Status
Vehicle(CarA)	Given
Behind(CarA, Ambulance)	Given
GreenSignal(Ambulance)	Derived

Therefore:

Proceed(CarA) is obtained.

Step 6 – Resolution Refutation

Step	Clauses Used	Result
4	C3 + C6 + C7 + GreenSignal(Ambulance)	Proceed(CarA)
5	Proceed(CarA) + C8	$\square$

The empty clause $\square$ is obtained.

Result

Proceed(CarA) is proved by Resolution.

Resolution Proof Tree

flowchart TD

A["Vehicle(Ambulance)"] --> C["ClearPath(Ambulance)"]

B["EmergencyType(Ambulance)"] --> C

C --> D["GreenSignal(Ambulance)"]

E["Vehicle(CarA)"] --> F["Proceed(CarA)"]

G["Behind(CarA, Ambulance)"] --> F

D --> F

F --> H["Resolve with $\neg \text{Proceed}(\text{CarA})$ "]

H --> I[" $\square$ Empty Clause"]

(c) Two Simultaneous Emergency Vehicles

The given rules can represent multiple vehicles because the rules contain variables. However, the present knowledge base does not specify priority or conflict handling when two emergency vehicles require the same path.

Additional Knowledge Required

Additional Rule / Fact	Purpose
EmergencyPriority(x,p)	Assign priority to an emergency vehicle
SameRoute(x,y)	Identify vehicles using the same route
HigherPriority(x,y)	Compare emergency priorities
AvailablePath(p)	Check path availability
AssignedPath(x,p)	Assign a path to a vehicle
Conflict(x,y)	Identify a routing conflict

Example rules:

$\text{HigherPriority}(x,y) \rightarrow \text{GiveWay}(y,x)$

$\text{SameRoute}(x,y) \wedge \text{HigherPriority}(x,y) \rightarrow \text{Hold}(y)$

Analysis

Situation	Current KB
One emergency vehicle	Can be handled
Multiple vehicles	Can be represented
Emergency priority	Not defined
Same-path conflict	Not defined
Alternative path selection	Not defined
Traffic coordination	Not defined

Result

The current KB is sufficient for the given CarA case. Additional **priority, path allocation and conflict-handling rules** are needed for two simultaneous emergency vehicles.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

Aim

To convert the given agricultural knowledge base into CNF and use Resolution Refutation to prove ApplyDripMethod.

(a) Conversion into CNF

P1

Given: SoilDry $\rightarrow$ IrrigationNeeded

Step	Conversion
Original	SoilDry $\rightarrow$ IrrigationNeeded
Eliminate implication	$\neg$ SoilDry $\vee$ IrrigationNeeded
CNF	$\neg$SoilDry $\vee$ IrrigationNeeded

P2

Given: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Step	Conversion
Original	IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
Eliminate implication	$\neg$ (IrrigationNeeded $\wedge$ CropWheat) $\vee$ ApplyDripMethod
Apply De Morgan's law	$\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod
CNF	$\neg$IrrigationNeeded $\vee$ $\neg$CropWheat $\vee$ ApplyDripMethod

P3

Given: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Step	Conversion
Original	$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk
Eliminate implication	$\neg$ ($\neg$ ApplyDripMethod $\wedge$ CropWheat) $\vee$ CropAtRisk
Apply De Morgan's law	ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk
CNF	ApplyDripMethod $\vee$ $\neg$CropWheat $\vee$ CropAtRisk

Facts

Fact	CNF
SoilDry = True	SoilDry
CropWheat = True	CropWheat

Final CNF

No.	Sentence	CNF
P1	SoilDry $\rightarrow$ IrrigationNeeded	$\neg$ SoilDry $\vee$ IrrigationNeeded
P2	IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod	$\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod
P3	$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk	ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk
F1	SoilDry = True	SoilDry
F2	CropWheat = True	CropWheat

(b) Resolution Refutation for ApplyDripMethod

Goal

ApplyDripMethod

Negated goal:

$\neg$ ApplyDripMethod

Add the negated goal to the knowledge base.

Resolution Steps

Step	Clauses Used	Resolvent
1	$\neg$ SoilDry $\vee$ IrrigationNeeded, SoilDry	IrrigationNeeded
2	$\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod, IrrigationNeeded	$\neg$ CropWheat $\vee$ ApplyDripMethod
3	$\neg$ CropWheat $\vee$ ApplyDripMethod, CropWheat	ApplyDripMethod
4	ApplyDripMethod, $\neg$ ApplyDripMethod	$\square$

Resolution Proof Tree

flowchart TD

A["SoilDry"] --> C[" $\neg$ SoilDry $\vee$ IrrigationNeeded"]

C --> D["IrrigationNeeded"]

$D \rightarrow F[\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}]$
 $B[\text{CropWheat}] \rightarrow F$
 $F \rightarrow G[\text{ApplyDripMethod}]$
 $G \rightarrow H[\neg \text{ApplyDripMethod}]$
 $H \rightarrow I[\square \text{ Empty Clause}]$

Result

ApplyDripMethod is proved by Resolution Refutation.

(c) Removing SoilDry

If SoilDry is removed, Rule P1 cannot be used to derive IrrigationNeeded.

Step	Before Removing SoilDry	After Removing SoilDry
1	SoilDry is available	SoilDry is unavailable
2	IrrigationNeeded is derived	IrrigationNeeded is not derived
3	CropWheat is available	CropWheat remains available
4	ApplyDripMethod is derived	ApplyDripMethod is not derived
5	$\neg$ ApplyDripMethod is not given	$\neg$ ApplyDripMethod is not given
6	CropAtRisk is not derived	CropAtRisk is not derived

P3 requires: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Although CropWheat is available, **$\neg$ ApplyDripMethod is not given.**

Therefore, P3 cannot be fired.

Result

CropAtRisk does not follow after removing SoilDry.

Also:

Conclusion	Status
IrrigationNeeded	Not derived
ApplyDripMethod	Not derived
CropAtRisk	Not derived

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

Aim

To construct the agent's belief state from the given percepts and use Modus Ponens and Forward Chaining to identify possible Wumpus and Pit locations and evaluate the safety of the given path.

(a) Belief State from Percepts

Given Percepts

Cell	Percept
[1,2]	Stench
[1,1]	Breeze
[2,2]	Glitter

Modus Ponens

Step	Given Fact	Rule Applied	Derived Fact
1	Stench[1,2]	$\text{Stench}[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$	$\text{WumpusAdjacent}[1,2]$
2	Breeze[1,1]	$\text{Breeze}[1,1] \rightarrow \text{PitAdjacent}[1,1]$	$\text{PitAdjacent}[1,1]$
3	Glitter[2,2]	$\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$	$\text{Gold}[2,2]$

Safe Cell Rule

$\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

No negative Wumpus or Pit facts are provided.

Rule	Required facts	Result
$\text{Safe}[x,y]$	$\neg \text{Wumpus}[x,y]$ and $\neg \text{Pit}[x,y]$	Cannot be derived

Belief State

Knowledge	Status
Stench[1,2]	Given
Breeze[1,1]	Given
Glitter[2,2]	Given

WumpusAdjacent[1,2]	Derived
PitAdjacent[1,1]	Derived
Gold[2,2]	Derived
Safe[x,y]	Not established

Result

The agent knows that there is a Wumpus adjacent to [1,2], a Pit adjacent to [1,1], and Gold at [2,2].

(b) Forward Chaining – Possible Locations

Using the usual four-neighbour adjacency:

Wumpus

Stench is present at [1,2].

Adjacent cells:

Source cell	Adjacent cells
[1,2]	[1,1], [1,3], [2,2]

Therefore, the possible Wumpus locations are:

[1,1], [1,3], [2,2]

Pit

Breeze is present at [1,1].

Adjacent cells:

Source cell	Adjacent cells
[1,1]	[1,2], [2,1]

Therefore, the possible Pit locations are:

[1,2], [2,1]

Forward Chaining

Step	Percept	Inference	Result
1	Stench[1,2]	Wumpus is adjacent	Wumpus: [1,1], [1,3], [2,2] possible
2	Breeze[1,1]	Pit is adjacent	Pit: [1,2], [2,1] possible
3	Glitter[2,2]	Gold is present	Gold at [2,2]

Reasoning Flow

flowchart TD

A["Stench at [1,2]"] --> B["Wumpus adjacent"]

B --> C["Possible Wumpus: [1,1], [1,3], [2,2]"]

D["Breeze at [1,1]"] --> E["Pit adjacent"]
 E --> F["Possible Pit: [1,2], [2,1]"]
 G["Glitter at [2,2]"] --> H["Gold at [2,2]"]

(c) Safety of the Given Path

Given path:

[1,1] → [1,2] → [2,2]

Cell	Knowledge	Decision
[1,1]	Possible Wumpus location	Not confirmed safe
[1,2]	Possible Pit location	Not confirmed safe
[2,2]	Gold is present, but Wumpus is possible	Not confirmed safe

Path Check

[1,1]

Stench at [1,2] makes [1,1] a possible Wumpus location.

[1,2]

Breeze at [1,1] makes [1,2] a possible Pit location.

[2,2]

Glitter confirms Gold at [2,2], but [2,2] is also a possible Wumpus location.

Result

The path [1,1] → [1,2] → [2,2] **cannot be confirmed as safe** using the available knowledge.

The agent can confirm the location of the gold, but it cannot confirm that all three cells in the path are free from danger.

